GRADE 10 · MATHEMATICS ENGLISH MEDIUM

QUESTION PAPER

Model Paper — 04

Third Term Test (Full Syllabus)

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Time: 3 hours **Total marks:** 100

Name: ..... Class: ..... Date: .....

Instructions:

- This question paper consists of two parts, **A** and **B**. Answer **all** the questions.
- Answer the 20 questions in **Part A** in the space provided on this paper itself. (2 marks for each question)
- Answer the 6 questions in **Part B** on the separate sheets of paper provided. (10 marks for each question)
- Show the steps of your calculations clearly. Draw graphs on the graph paper provided.
- Where necessary, take the value of π as $\frac{22}{7}$. The use of calculators is not allowed.

Part	A	B1	B2	B3	B4	B5	B6	Total
Marks	/40	/10	/10	/10	/10	/10	/10	/100

Part A

Answer all the questions in the space provided on this paper. (2 × 20 = 40 marks)

1. Find the square root of 3249 using the division method.

Answer: .....

2. Simplify: $3\frac{3}{4} - 1\frac{2}{3} \times 1\frac{1}{5}$

Answer: .....

3. Expand: $(3x + 2y)^2$

Answer: .....

4. Factorise: $2x^2 + 7x - 15$

Answer: .....

5. Find the least common multiple (L.C.M.) of the algebraic expressions $8x^2y$, $12xy^2$ and $6x^3$.

Answer: .....

6. Simplify: $\frac{a}{a-b} + \frac{b}{b-a}$

Answer: .....

7. The area of a sector of radius 7 cm is 38.5 cm^2. Find the angle at the centre of the sector.

Answer: .....

8. In triangle ABC, $\hat{A} = x°$ and $\hat{B} = 2x°$. If the exterior angle at vertex C is 117°, find the magnitude of $\hat{B}$.

Answer: .....

9. Find the value of: $\log_5 125 - \log_2 \frac{1}{8}$

Answer: .....

10. Find the value of: $\lg 4 + 2 \lg 5$

Answer: .....

11. In an arithmetic progression, the 5th term is 17 and the 10th term is 32. Find the common difference.

Answer: .....

12. If ξ = {integers from 1 to 12} and A = {factors of 12}, find n(A′).

Answer: .....

13. An article bought for Rs. 4 800 was sold at a loss of 15%. Find the selling price.

Answer: .....

14. 8 identical pumps take 9 hours to empty a tank. Find the time taken by 6 such pumps to empty the tank.

Answer: .....

15. Make r the subject of the formula $A = \pi r^2$. $(r > 0)$

Answer: .....

16. Solve the inequality $3(x - 1) > 2x + 4$ and write the least integer value that x can take.

Answer: .....

17. In a pie chart, a sector with an angle of 54° at the centre represents 27 students. Find the total number of students represented by the pie chart.

Answer: .....

18. Find the probability of getting a number greater than 4 when a fair die is rolled once.

Answer: .....

19. In the parallelogram PQRS, PQ = 8 cm and the area is 48 cm^2. Find the perpendicular distance between PQ and SR.

Answer: .....

20. The volume of a cuboid is 360 cm^3. The length of its base is 9 cm and the width is 8 cm. Find its height.

Answer: .....

Part B

Answer all the questions on separate sheets of paper. (10 × 6 = 60 marks)

1. (a) Solve: $2x - y = 7$, $3x + 2y = 21$ (4 marks)

(b) Solve by factorising: $x^2 + 2x - 24 = 0$ (2 marks)

(c) The sum of a number and its square is 42.

(i) Taking the number as x, form a quadratic equation in x. (1 mark)

(ii) Solve that equation and find the possible values of the number. (2 marks)

(iii) If the number is positive, find the value of $2x - 1$. (1 mark)

2. An incomplete table of values prepared for drawing the graph of the function $y = 6 - 2x^2$ is given below.

x	−3	−2	−1	0	1	2	3
y	−12	−2	.....	6	4	−2	.....

(a) Complete the blanks in the table. (1 mark)

(b) Draw the graph taking 10 small squares along the x-axis to represent 1 unit and 10 small squares along the y-axis to represent 2 units. (3 marks)

(c) Using the graph,

(i) write the maximum value of the function and the coordinates of the turning point. (1 mark)

(ii) write the approximate roots of the equation $6 - 2x^2 = 0$. (2 marks)

(iii) write the range of values of x for which y is not negative. (1 mark)

(iv) write the value of y when $x = -2.5$. (1 mark)

(d) Write the function of the graph obtained when the graph is translated 6 units downwards along the y-axis. (1 mark)

3. A cylindrical metal pipe has an outer radius of 7 cm, an inner radius of 3.5 cm and a length of 20 cm.

(a) Find the volume of metal used to make the pipe. (3 marks)

(b) Find the area of the outer curved surface of the pipe. (2 marks)

(c) If the mass of 1 cm^3 of the metal is 8 g, find the mass of the pipe in kg. (2 marks)

(d) When one end of the pipe is closed and it is filled with water, find the volume of water it can hold, in litres. How many such filled pipes of water are needed to fill a bucket of 7.7 litres? (3 marks)

4. In triangle ABC, AB = AC. BE is the perpendicular drawn from B to AC and CF is the perpendicular drawn from C to AB. BE and CF intersect at O.

(a) Draw a rough sketch including the above information. (1 mark)

(b) Prove that triangle BFC and triangle CEB are congruent. (3 marks)

(c) Show that BE = CF. (1 mark)

(d) Show that OB = OC. (2 marks)

(e) If $B\hat{A}C = 50°$, find the magnitude of $B\hat{O}C$. (3 marks)

5. The table below shows the time (in minutes) taken by 50 students to solve a puzzle. (0 – 10 means 0 minutes or more but less than 10 minutes.)

Time (minutes)	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
Number of students	6	9	15	12	8

(a) Write the modal class. (1 mark)

(b) Using the class mid-values, find the mean time taken by a student. (4 marks)

(c) Find the percentage of students who took less than 20 minutes. (2 marks)

(d) If each of the 8 students in the 40 – 50 class had taken 10 minutes less, find the new mean time of the 50 students. (3 marks)

6. In a survey of 60 people, it was found that 35 read newspaper P, 28 read newspaper Q, and 10 read both newspapers.

(a) Represent this information in a Venn diagram. (3 marks)

(b) Find the number of people who read at least one newspaper. (2 marks)

(c) Write the number of people who read neither newspaper. (1 mark)

(d) Find the probability that a person selected at random

(i) reads only one newspaper, (2 marks)

(ii) reads only newspaper Q. (2 marks)

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